

Q.1

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TE|IT|V

Syn.

a) What is the need of system call? Following code will create how many process and why show process created with neat diagram. (5)

```
int main()
{
    int i;
    for(i=0; i<3; i++)
        fork();
    return(0);
}
```

Answer:

When a program in user mode requires access to RAM or a hardware resource, it must ask the kernel to provide access to that resource. This is done via something called a **system call**.

formula: $((2^n)-1)+1$, where $((2^n)-1)$ are the child process and 1 is for parent process.

Why: Fork: fork is a system call used to create a new process. Fork creates exact copy of the parent process except process id.

Here fork is called 3 times. So n is 3. By putting $n=3$ in the above formula, we get no. Of process created=8, among which 7 are the child process and 1 is the parent process.

Points:

1. for what is need of system call .
1. for why
1. for formula
1. for calculations.
1. for diagram showing for fork at respective level.

b) What are the advantages and disadvantages of layered architecture of OS ? (5)

Advantages : 1. Layered architecture has all the advantages of modular design.

2. In modular design, the system is divided into several modules and each module is designed independently .

3. Each layer can be designed, coded and tested independently.

4. Layered approach simplifies the design, specification and implementation of an O.S.

5. Drawback: The OS function must be carefully assigned to various layers because a layer can make use only of the functionality provided by the layer beneath it.

Points:

4. for advantages
1. for disadvantage

c) What is RAID? Why it is needed? Explain RAID 10 with neat diagram along with its advantage and disadvantages. (10)

Answer: RAID: Redundant array of independent disks.

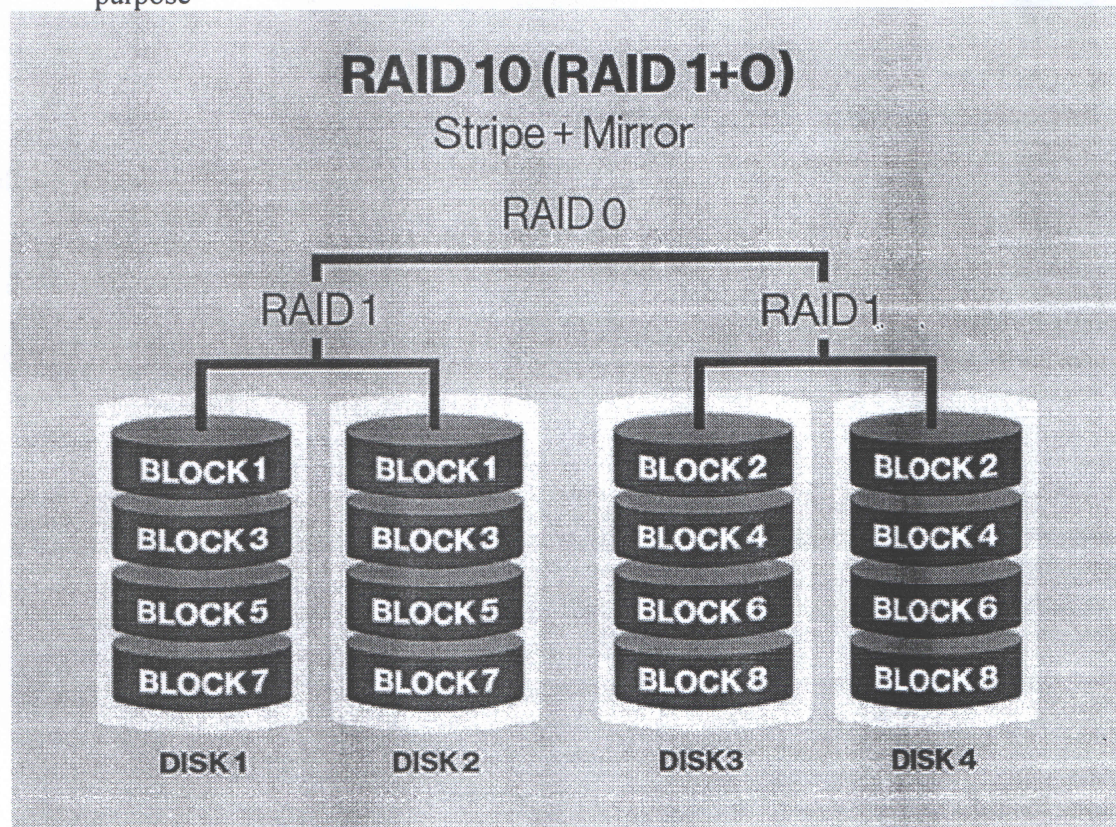
Need:

1. The rate in improvement in secondary storage performance has been considerably less than the rate for processors and main memory.
2. So additional gains in performance are to be had by using multiple parallel components.
3. In the case of disk storage, this leads to the development of arrays of disks that operate independently and in parallel.
4. With multiple disks, separate I/O requests can be handled in parallel, as long as the data required reside on separate disks.

5. Further, a single I/O request can be executed in parallel if the block of data to be accessed is distributed across multiple disks.

RAID 10:

1. it is called as mirror striped. 1st stripping takes place then mirroring.
2. It requires minimum 4 hard disks.
3. It provides fault tolerance of RAID 1 and speed of RAID 0.
4. It actually uses only 50% of the available disks space because 50% is used for mirroring purpose



points:

1. for what is RAID.
4. for its need
4. for what is RAID 10.
1. for RAID 10 diagram.

Q.2.a) What resources are used when a thread is created? How do they differ from those used when a process is created? Differentiate between user level thread and kernel level thread for 5 different points (10).

Answer: 1. Because a thread is smaller than a process, thread creation typically uses fewer resources than process creation.

2. Creating a process requires allocating a process control block (PCB), a rather large data structure.
3. The PCB includes a memory map, list of open files, and environment variables.
 4. Allocating and managing the memory map is typically the most time consuming activity.
 5. Creating either a user or kernel thread involves allocating a small data structure to hold a register set, stack, and priority.

USER LEVEL THREAD

User threads are implemented by users.
OS doesn't recognize user level threads.
Implementation of User threads is easy.
Context switch time is less.
Context switch requires no hardware support.
If one user level thread performs a blocking operation then entire process will be blocked.
Example : Java thread, POSIX threads.

KERNEL LEVEL THREAD

kernel threads are implemented by OS.
Kernel threads are recognized by OS.
Implementation of Kernel thread is complicated.
Context switch time is more.
Hardware support is needed.
If one kernel thread performs a blocking operation then another thread can continue execution.
Example : Windows Solaris.

Points:

1. for thread process comparison
1. for pcb
2. for pcb contents and its overhead.
1. for data structure for thread creation
5. for 5 differences.

Or

Q.2 a) Explain process state transition diagram, also write down in each state what will be the contents of PCB with respect to the state of the process? Why process state diagram exists?

- As a process executes, it changes state. The state of a process is defined in part by the current activity of that process.

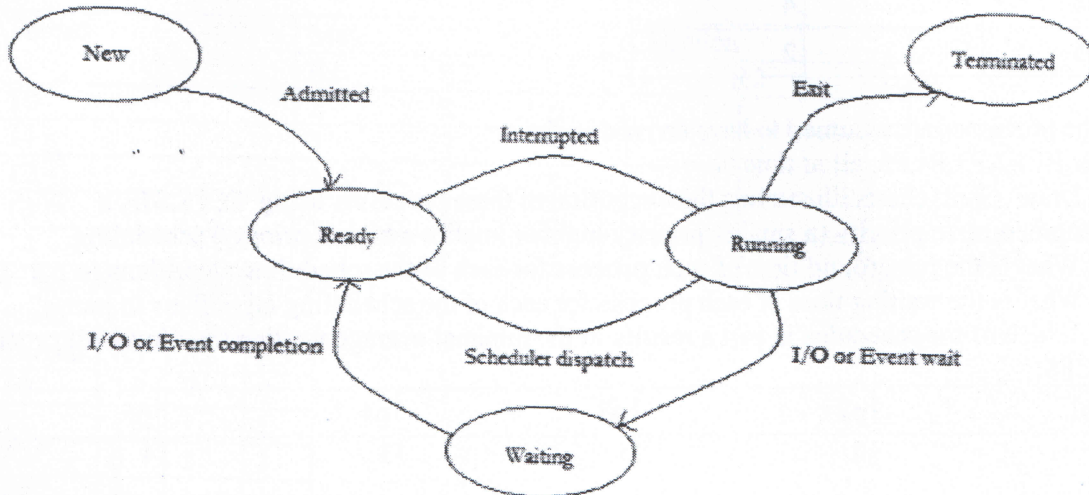


Fig: Process state transition diagram.

- Process can have one of the following five states at a time.
- 1. **New state:** A process that just has been created but has not yet been admitted to the pool of execution processes by the operating system. Every new operation which is requested to the system is known as the new born process.
- 2. **Ready state:** When the process is ready to execute but he is waiting for the CPU to execute then this is called as the ready state. After completion of the input and output the process will be on ready state means the process will wait for the processor to execute. **PCB will contain the state of the process as ready.**
- 3. **Running state:** The process that is currently being executed. When the process is running under the CPU, or when the program is executed by the CPU, then this is called as the running process and when a process is running then this will also provide us some outputs on the screen. **PCB will contain the state of process as running.**
- 4. **Waiting or blocked state:** A process that cannot execute until some event occurs or an I/O completion. When a process is waiting for some input and output operations then this is called as the waiting state and in this process is not under the execution instead the process is stored out of memory and when the user will provide the input and then this will again be on ready state. **PCB will contain the state of process as blocked.**
- 5. **Terminated state:** After the completion of the process, the process will be automatically terminated by the CPU. So this is also called as the terminated state of the process. After executing the complete process the processor will also deallocate the memory which is allocated to the process. So this is called as the terminated process.

Why process state diagram exists:

1. Program execution contains alternate CPU and i/o phase.
2. Program execution always starts with CPU phase and
3. It always terminates with CPU phase.
4. In between alternate CPU, i/o phase. This leads to process state transition diagram.

Points:

- 1 for process state transition diagram.
5. for state in transition diagram.
4. for why state diagram exists.

Q.2 b) Consider the following set of processes, with the length of the CPU burst time given in milliseconds: (10)

Process	Burst Time	Priority
p1	10	3
p2	1	1
p3	2	3
p4	1	4
p5	5	2

The processes are assumed to have arrived in the order P1,P2,P3,P4,P5, all at time 0.

- a. Draw Gantt charts illustrating the execution of these processes using FCFS, SJF, a nonpreemptive priority (a smaller priority number implies a higher priority) scheduling.
- b. What is the turnaround time of each process for each of the scheduling algorithms in part a?
- c. What is the waiting time of each process for each of the scheduling algorithms in part a?
- d. Which of the schedules in part a results in the minimal average waiting time (over all processes)?

FCFS:

p1	p2	p3	p4	p5	
0	10	11	13	14	19

SJF:

p2	p4	p3	p5	p1	
0	1	2	4	9	19

Np priority

p2	p5	p1	p3	p4	
0	1	6	16	18	19

Turnaround time

Process	FCFS	SJF	Priority
p1	10	11	16
p2	11	1	1
p3	13	4	18
p4	14	2	19
p5	19	9	6

c. Waiting time (turnaround time minus burst time)

Process	FCFS	SJF	Priority
p1	0	9	6
p2	10	0	0
p3	13	1	18
p4	14	2	19
p5	14	4	1

Average wait time:

FCFS=51/5=10.2

SJF=16/5=3.2

Priority=44/5=8.8

minimal average waiting time (over all processes): Shortest Job First

Points:

1. for gantt chart
1. for avg. waiting time of each process
1. for avg. turnaround time of each process
1. for minimal average waiting time over all process.

Q.3

a) An operating system has 3 types of resources with the following quantities of each type:

Resource Type: R1 R2 R3

Quantity of that type: 13 8 10.

Show the calculations of the Banker's algorithm to determine whether or not this system is in a safe state. (10)

Thread	Max			Allocation		
	R1	R2	R3	R1	R2	R3
T0	2	8	0	1	1	0
T1	5	2	2	4	1	1
T2	4	3	9	1	1	7
T3	2	2	3	2	1	1
T4	2	1	3	2	0	0

Available: $13-10, 8-4, 10-9=341$.

Need:

Thread	R1	R2	R3
T0	1	7	0
T1	1	1	1
T2	3	2	2
T3	0	1	1
T4	0	1	3

Safe state sequence : T2: 752, T3: 869, T4: 10710, T5: 12710, T1: 13810.

Points:

2. for calculating need.
2. for calculating available.
5. for determining which Thread can run / can not run and why?
1. for showing safe sequence if any.

Or

What is 'wait- for' graph scheme? What is its drawback. Explain deadlock detection algorithm. (10).

Ans - Wait for graph is a resource allocation graph where an edge p_i to p_j implies process p_i is waiting for process p_j to release a resource that a process p_i needs. Process p_j waits for process p_k . A deadlock exists in the system if and only if the wait-for graph contains a cycle.

The wait-for graph scheme is not applicable to a resource-allocation system with multiple instances of each resource type.

Deadlock detection algorithm that is applicable to such a system is as follows.

1. Data structure required

Available. A vector of length m indicates the number of available resources of each type.

• Allocation. An $n \times m$ matrix defines the number of resources of each type currently allocated to each process.

• Request. An $n \times m$ matrix indicates the current request of each process. If $\text{Request}[i][j]$ equals k , then process P_i is requesting k more instances of resource type R_j .

Algorithm

1. Let Work and Finish be vectors of length m and n , respectively. Initialize Work $\leftarrow$ Available. For $i = 0, 1, \dots, n-1$, if $\text{Allocation}[i] \neq 0$, then $\text{Finish}[i] \leftarrow \text{false}$; otherwise, $\text{Finish}[i] \leftarrow \text{true}$.

2. Find an index i such that both

a. $\text{Finish}[i] \leftarrow \text{false}$

b. $\text{Request}[i] \leq \text{Work}$

If no such i exists, go to step 4.

3. $\text{Work} \leftarrow \text{Work} + \text{Allocation}[i]$;

$\text{Finish}[i] \leftarrow \text{true}$

Go to step 2.

4. If $\text{Finish}[i] = \text{false}$, for some i , $0 \leq i < n$, then the system is in a deadlocked state.

Moreover, if $\text{Finish}[i] = \text{false}$, then process P_i is deadlocked.

Points:

2. for wait for graph.

1. its disadvantage

7. for deadlock detection algorithm.

b) Differentiate between Monitor and Semaphore for 4 different points and their use. (5)

Basis for Comparison	Semaphore	Monitor
Basic	Semaphores is an integer variable S.	Monitor is an abstract data type.
Action	The value of Semaphore S indicates the number of shared resources available in the system	The Monitor type contains shared variables and the set of procedures that operate on the shared variable.
Access	When any process access the shared resources it perform wait() operation on S and when it releases the shared resources it performs signal() operation on S.	When any process wants to access the shared variables in the monitor, it needs to access it through the procedures.
Condition variable	Semaphore does not have condition variables. Monitor has condition variables.	

Monitor and Semaphore are constructs used to provide synchronization among concurrent process

Points:

4. for 4 differences

1. for use

c) What does the positive and negative value of counting semaphore indicates? A counting semaphore was initialized to 10. Then 6 P (wait) operations and 4V (signal) operations were completed on this semaphore. The resulting value of the semaphore is----- (5)

(a) 0 (b) 8 (c) 10 (d) 12

Ans: option(b).

6P $\Rightarrow$ decrements the semaphore 6 times. Hence, the value becomes 4.

4V $\Rightarrow$ increments the semaphore 4 times. Hence, the value becomes 8.

The positive value of counting semaphore indicates that those many down (P) operations can be carried out successfully. The negative value of counting semaphore indicates that the number of blocked processes.

Points:

2. for +ve and -ve value of semaphore

1. for choosing correct option

2. for explanation.

Q.4

a)

1.) Consider a system with 80% hit ratio, 50 nano-seconds time to search the associative registers, 750 nano seconds time to access memory. Find the time to access a page. (5)

a. When the page number is in associative memory.

b. When the time to access a page when not in associative memory.

c. Find the effective memory access time.

Ans:

a. The time required is 50 nano seconds to get the page number from associative memory and 750 nano-seconds to read the desired word from memory.

Time = $50 + 750 = 800$ nano seconds.

b.

Now the time when not in associative memory is

Time = $50 + 750 + 750 = 1550$ nano seconds

One memory access extra is required to read the page table from memory.

c.

Effective access time = Page number in associative memory +

Page number not in associated memory.

Page number in associative memory = $0.8 * 800$.

Page number not in associated memory = $0.2 * 1550$.

Effective access time = $0.8 * 800 + 0.2 * 1550 = 950$ nano seconds.

points

1. for a

1. for b

1. for formula in c

2. for calculations in c.

2.) What is the disadvantage of paging. ? Consider a logical address space of 8 pages of 1024 words mapped into memory of 32 frames.

a. How many bits are there in the logical address ?

b. How many bits are there in physical address ?

Ans:

Paging disadvantage: internal fragmentation in last page, modularity of program not kept.

a. Logical address will have

3 bits to specify the page number (for 8 pages) .

10 bits to specify the offset into each page ($2^{10} = 1024$ words) = 13 bits.

b. For (2^5) = 32 frames of 1024 words each (Page size = Frame size) We have $5 + 10 = 15$ bits.

Points:

1. for disadvantage of paging

2. for a.

2. for b.

Or

a) Main memory 64MB, logical address is 32 bits, page size is 4KB. Compute total space wasted in maintain page table by the system. (10)

Answer: Main memory = 64MB = 64 M locations = 2^{26} location. Therefore physical address contains 26 bit.

Page size is 4KB. As memory is byte addressable, page contains 4K entries = $2^{10} * 2^2 = 2^{12}$.

Therefore 12 bit in logical address for offset, 12 bit in physical address are for offset.

Therefore for frame number we have = $26 - 12 = 14$ bit.

For page number we have = $32 - 12 = 20$ bit.

Memory is byte addressable 14b = 16bit = 2 Byte

Therefore size of each entry in main memory is 2 byte.

Total entries are 2^{20} .

Therefore memory wasted = $1M * 2B = 2MB$.

Points:

2. for calculating locations in MM.

2. for calculating offset.

2. for calculating frame number.

2. for calculating size of each entry in MM.

2. for calculating memory wasted in maintaining page table.

b) If the reference string is 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, take 3 page frames. Solve using LRU and Optimal page replacement policy, calculate page fault ratio. (10)

LRU:

7	0	1	2	0	3	0	4	2	3	0	3
7	7	7	2	2	2	2	4	4	4	0	0
	0	0	0	0	0	0	0	0	3	3	3
		1	1	1	3	3	3	2	2	2	2
f	f	f	f		f		f	f	f	f	

Page fault 9

page fault ratio: $(9/12) * 100 = 75\%$

Optimal

7	0	1	2	0	3	0	4	2	3	0	3
7	7	7	2	2	2	2	2	2	2	0	0
	0	0	0	0	0	0	4	4	4	4	4
	1	1	1	3	3	3	3	3	3	3	3
f	f	f	f		f		f		f		

Page fault 7

page fault ratio: $(7/12) \times 100 = 58.33\%$

points:

1. for page fault

4. for showing page frame entry

Q.5

a) Cylinder requests are 55, 58, 39, 18, 90, 160, 150, 38, 184. There are 200 cylinders numbered from 0 – 199. The disk head starts at number 100. Calculate total head movement and average seek length using SCAN and CSCAN disk scheduling algorithm. Assume head moves towards increasing track number. (10)

Answer:

SCAN:

$(100-150) + (150-160) + (160-184) + (184-90) + (90-58) + (58-55) + (55-39) + (39-38) + (38-18) = 250$

Avg seek length = $250/9 = 27.7$

CSCAN:

$(100-150) + (150-160) + (160-184) + (184-18) + (18-38) + (38-39) + (39-55) + (55-58) + (58-90) = 322$

Avg. seek length = $322/9 = 35.7$

points:

1. for graph

3. for calculating total head movement

1. for average seek length

b) Explain contiguous file allocation method example with neat diagram with its advantages and disadvantages. (10)

Contiguous Allocation

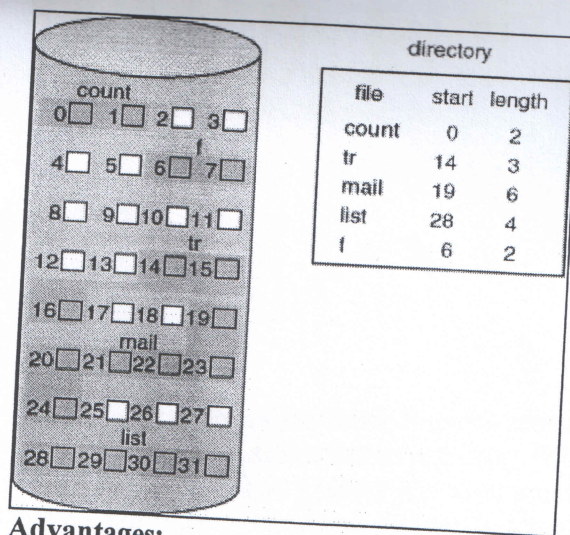
1. In this scheme, each file occupies a contiguous set of blocks on the disk.

2. For example, if a file requires n blocks and is given a block b as the starting location, then the blocks assigned to the file will be: $b, b+1, b+2, \dots, b+n-1$. This means that given the starting block address and the length of the file (in terms of blocks required), we can determine the blocks occupied by the file.

3. The directory entry for a file with contiguous allocation contains

- Address of starting block
- Length of the allocated portion.

4. The file 'mail' in the following figure starts from the block 19 with length = 6 blocks. Therefore, it occupies 19, 20, 21, 22, 23, 24 blocks.



Advantages:

- Both the Sequential and Direct Accesses are supported by this. For direct access, the address of the k th block of the file which starts at block b can easily be obtained as $(b+k)$.
- This is extremely fast since the number of seeks are minimal because of contiguous allocation of file blocks.

Disadvantages:

- This method suffers from both internal and external fragmentation. This makes it inefficient in terms of memory utilization.
- Increasing file size is difficult because it depends on the availability of contiguous memory at a particular instance.

Points:

3. for contiguous file allocation

1. for example

2. for diagram

2. for advantage and

2. for disadvantage.

Or

Explain three methods of record blocking in file management with neat diagrams. (10).

Answer:

Fixed blocking: Fixed-length records are used, and an integral number of records are stored in a block.

Disadvantage: There may be unused space at the end of each block. This is referred to as internal fragmentation.

Variable-length spanned blocking: Variable-length records are used and are packed into blocks with no unused space. Thus, some records must span two blocks, with the continuation indicated by a pointer to the successor block.

Advantage: Variable-length spanned blocking is efficient of storage and does not limit the size of records.

Disadvantage: difficult to implement. Records that span two blocks require two I/O operations, and files are difficult to update, regardless of the organization

Variable-length unspanned blocking: Variable-length records are used, but spanning is not employed.

Disadvantage: There is wasted space in most blocks because of the inability to use the remainder of a block if the next record is larger than the remaining unused space and limits record size to the size of a block.

Points:

3. for fixed length records

4. for variable length spanned record

3. for variable length unspanned record.